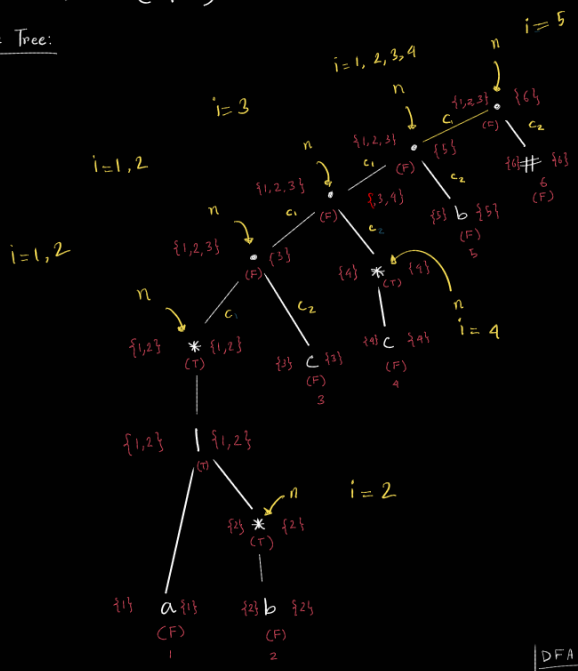
Answer to the question no 01

Given Regular expression: $(a|b^*)^* c c^* b$

Augmented form: $(a|b^*)^* c c^* b \#$

Syntax Tree:



I might have added something extra in the syntax tree sorry i forgot, what extra things I added.

Followpos:

Node	Followpos
1	{1,2,3}
2	{1,2,3}
3	{4,5}
4	{4,5}
5	{6}
6	

State Table:

State	a	b	c
A	A	A	B
B	D	C	B
C	D	D	D
D	D	D	D

Dtrans [A,a] = followpos (1)
= {1,2,3}
= A

Dtrans [A,b] = followpos (2)
= {1,2,3}
= A

Dtrans [A,c] = followpos (3)
= {4,5} → B

Dtrans [B,a] = { } → D

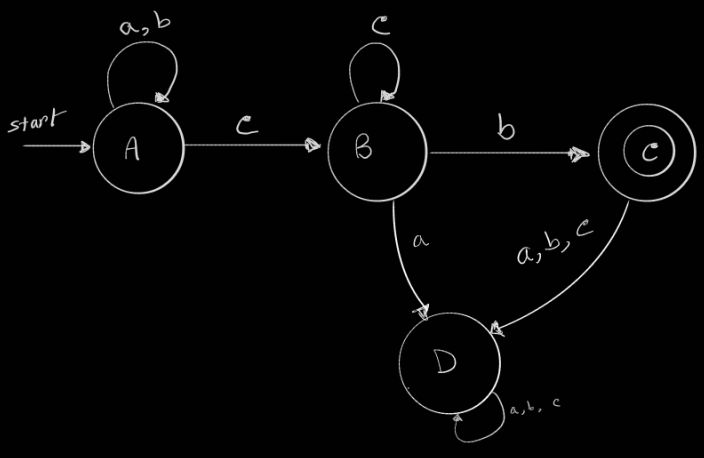
Dtrans [B,b] = followpos (5)
= {6} → C

Dtrans [B,c] = followpos (4)
= {4,5}
= B

Dtrans [C,a] = { } → D

Dtrans [C,b] = { } → D

Dtrans [C,c] = { } → D

{1,2,3} → A
{4,5} → B
{6} → C ∴ the end node
{ } → D

DFA: